

$$V = \frac{|ax + by + cz + d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$1) M = [-7; 0; -1]$$

$$\rho: 4x + 12y - 3z - 1 = 0$$

$$V = \frac{|4 \cdot (-7) + 12 \cdot 0 - 3 \cdot (-1) - 1|}{\sqrt{4^2 + 12^2 + (-3)^2}} = \frac{|-28 + 3 - 1|}{\sqrt{16 + 144 + 9}} = \frac{|-26|}{\sqrt{169}} = \frac{26}{13} = \underline{\underline{2}}$$

$$2) M = [-7; 3; -1]$$

$$\rho = \overleftrightarrow{ABC}$$

$$A = [1; 0; 1]$$

$$B = [2; 2; 1]$$

$$C = [0; 0; 2]$$

$$\overrightarrow{AB} = (1; 2; 0)$$

$$\overrightarrow{AC} = (-1; 0; 1)$$

$$x = 1 + t - s$$

$$y = 2t \Rightarrow t = \frac{y}{2}$$

$$z = 1 + s \Rightarrow s = z - 1$$

$$x = 1 + \frac{y}{2} - (z - 1)$$

$$x = 1 + \frac{y}{2} - z + 1 \quad | \cdot 2$$

$$2x = 2 + y - 2z + 2$$

$$0 = -2x + y - 2z + 4 \quad | \cdot (-1)$$

$$0 = 2x - y + 2z - 4$$

$$V = \frac{|-14 - 3 - 2 - 4|}{\sqrt{4 + 1 + 4}} = \frac{|-23|}{\sqrt{9}} = \frac{23}{3}$$

Postup:

$$1) \mathcal{P}: ax+by+cz+d=0$$

$$\vec{n}_P = \vec{s}_P \wedge M \in \mathcal{P}$$

$$2) P \in \mathcal{P} \cap \mathcal{P}$$

$$3) V = |MP| = \sqrt{(\quad)^2 + (\quad)^2 + (\quad)^2}$$

$$1) M = [0; 1; -4]$$

$$P: x = 2+t$$

$$y = 3$$

$$z = 2-t$$

$$\vec{s}_P = (1; 0; -1)$$

$$\mathcal{P}: x - z + d = 0$$

$$M \in \mathcal{P}: 0 - (-4) + d = 0$$

$$4 + d = 0$$

$$d = -4$$

$$\mathcal{P}: x - z - 4 = 0$$

$$2+t - (2-t) - 4 = 0$$

$$2+t - 2+t - 4 = 0$$

$$2t - 4 = 0$$

$$2t = 4$$

$$t = 2$$

$$x = 2+2 = 4$$

$$z = 2-2 = 0$$

$$P = [4; 3; 0]$$

$$V = |\vec{MP}| = \sqrt{16+4+16} = \sqrt{36} = \underline{\underline{6}}$$

$$2) M = [-2; 1; 0]$$

$$\mathcal{P}: x = 1-2t$$

$$y = 2+t$$

$$z = 3t$$

$$1) \mathcal{P}: ax+by+cz+d=0$$

$$\vec{n}_P = \vec{s}_P \wedge M \in \mathcal{P}$$

$$2) P \in \mathcal{P} \cap \mathcal{P}$$

$$3) V = |MP| = \sqrt{(\quad)^2 + (\quad)^2 + (\quad)^2}$$

$$-2x + y + 3z + d = 0$$

$$4 + 1 + d = 0$$

$$d = -5$$

$$-2x + y + 3z - 5 = 0$$

$$-2(1-2t) + 2+t + 9t - 5 = 0$$

$$-2 + 4t + 2 + t + 9t - 5 = 0$$

$$14t - 5 = 0$$

$$14t = 5$$

$$t = 5/14$$

$$x = 1 - 2 \cdot \frac{5}{14} = \frac{2}{7}$$

$$y = 2 + \frac{5}{14} = \frac{33}{14}$$

$$z = 3 \cdot \frac{5}{14} = \frac{15}{14}$$

$$P = [\frac{2}{7}; \frac{33}{14}; \frac{15}{14}] \dots$$

$$3) M = [1; 0; 5]$$

$$P = \overleftrightarrow{AB}$$

$$\begin{matrix} A = [0; 1; 0] \\ B = [1; 0; 2] \end{matrix} \} \overrightarrow{AB} = (1; -1; 2) = \vec{s}_P = \vec{h}_g$$

$$P: x = t = 2 \quad P = [2; -1; 4]$$

$$y = 1 - t = -1 \quad V = |MP| = \sqrt{1+1+1} = \underline{\underline{\sqrt{3}}}$$

$$z = 2t = 4$$

$$\rho: x - y + 2z + d = 0$$

$$1 \quad +10 \quad +d = 0$$

$$\underline{\underline{d = -11}}$$

$$t - 1 + t + 4t - 11 = 0$$

$$6t - 12 = 0$$

$$-12 = -6t$$

$$\underline{\underline{t = 2}}$$